

DIGITAL PRODUCT DESIGN

Lecture 1: Introduction and Course Overview

Alex Moehring

Classroom Logistics

- Participation:
 - Press and hold microphone for in class participants
 - Virtual participants raise your zoom hand
 - Chat will not be monitored

Attendance

- Classroom has limited capacity and you have heterogeneous preferences
- I will send a Brightspace survey asking your preferred modality
- I have to give MBT students priority given this is a required course
 - I will randomly allocate remaining in-person seats to those wanting to attend in person
- Goal is to allocate spots in classroom to those who prefer to attend in person

Outline

- Motivation
 - What is different about digital products?
- Overview of course, what course is and is not
- Details of course
- Introductions

INTRODUCTION TO DIGITAL PRODUCTS

Something Has Changed in the Economy

Firm	Year Founded	Employees	Mkt Cap
BMW	1916	116,000	\$53B
Uber	2009	7,000	\$78B
Mariott	1927	200,000	\$17B
AirBNB	2008	5,000	\$31B
Walt Disney	1923	185,000	\$260B
Facebook	2004	12,691	\$783B

Something Has Changed in the Economy, Part 2

Something Has Changed in the Economy, Part 3

Radio Shack
AMERICA'S TECHNOLOGY STORE

PRESIDENTS' BIRTHDAY SALE!
DON'T DELAY! 3-DAY SPECIALS ABOVE GOOD SATURDAY THRU MONDAY ONLY!

0% INTEREST!
NO PAYMENTS UNTIL MAY!
NO DOWN PAYMENT!
HURRY! OFFER ENDS TUESDAY FEBRUARY 19

COME IN AND TAKE ADVANTAGE OF THESE OTHER FANTASTIC VALUES!

INTRODUCTORY SPECIAL!
Tandy 1000 TL/3 Computer System
Save \$670
\$1599
Low As \$99 Per Month. Reg. \$229.99
Includes:
• 286-Based PC
• Color Monitor
• 20MB SmartDrive™ Hard Drive
• Easy-to-Use 10-in-1 DeskMate Software
• Bonus Package For DeskMate
• DeskMate C&A Write
• Quickset
• 2-Button Mouse

VHS Camcorder
Save \$100
\$799
Low As \$69 Per Month. Reg. \$899.99
Rechargeable Battery. 300 minutes video. Light for instant shooting. 2-lens zoom. Auto-contrast. Auto-iris.

3-Way Speaker With Massive 15" Woofer
Save \$110
\$149.95
Low As \$19.95 Per Month. Reg. \$259.95
Optimum Music Reproduction. 40 watts. 15" woofer. 3" tweeter. 3" midrange. Auto-contrast. Auto-iris.

All-Weather Stereo
Cut 34%
\$1188
Realistic STEREO MATP-100 FM personal receiver shrugs off rain, wind, and noise. 100 watts. 100% duty cycle.

AM/FM Clock Radio
30% Off
\$1388
Chromatone-201 clock radio's compact size cuts weight. 100 watts. 100% duty cycle. 100% duty cycle.

In-Ear Stereo Phones
HALF PRICE!
\$788
Realistic in-ear phones weigh just 0.8 ounces. 100% duty cycle. 100% duty cycle.

Micro-Thin™ Calculator
39% Off
\$488
Radio Shack EC-413 is around the size of a credit card. Solar powered. 100% duty cycle. 100% duty cycle.

Mobile Cellular Telephone
Save \$100
\$199
Low As \$19 Per Month. Reg. \$299.99
Includes:
• Mobile Cellular Telephone
• Mobile Cellular Telephone
• Mobile Cellular Telephone

Deluxe Portable CD Player
Save \$40
\$159.95
Low As \$19 Per Month. Reg. \$199.95
Realistic CD-350 has 16-sonar memory. Headphones extra. 100% duty cycle. 100% duty cycle.

Tiny Dual-Superhet Radar Detector
Save \$60
\$79.95
Low As \$7.99 Per Month. Reg. \$139.95
Realistic TRD-300 lets you drive with confidence. Separate X and K band tones. 100% duty cycle. 100% duty cycle.

Compact 10-Channel Desktop Scanner
Save \$30
\$99.95
Low As \$9.99 Per Month. Reg. \$129.95
Realistic PRD-37 lets you catch the news as it happens. Hear police, fire, taxi, military, VHS more. 100% duty cycle. 100% duty cycle.

Mobile CB With Channel Controls on Mike
HALF PRICE!
\$49.95
Low As \$4.99 Per Month. Reg. \$99.95
Realistic TRC-430 lets you get highway info or help. you'll never have to drive "alone". 100% duty cycle. 100% duty cycle.

Our Easiest-to-Use Phone Answerer
Cut 17%
\$49.95
Low As \$4.99 Per Month. Reg. \$59.95
DUALFORM™ 160-241 answerer is ready to use—just plug it in. 100% duty cycle. 100% duty cycle.

Handheld Voice-Activated Cassette Tape Recorder
40% Off
\$29.95
Low As \$2.99 Per Month. Reg. \$49.95
Realistic CTM-85 makes an excellent "recorder" for students. 100% duty cycle. 100% duty cycle.

20-Memory Speed-Dial Phone
Cut 33%
\$29.95
Low As \$2.99 Per Month. Reg. \$44.95
Realistic MDP-200. 20-number speed-dial. 100% duty cycle. 100% duty cycle.

Check Your Phone Book for the Radio Shack Store or Dealer Nearest You.
Most Major Credit Cards Welcome.



Something Has Changed in the Economy, Part 4

\$1.2 million worth of applications in a smartphone today (2018 update)

Application	\$ 2018	Original Device Name	Year*	MSRP	\$ 2018
1 Video conferencing	Free	Compression Labs VC	1982	\$250,000	\$639,039
2 GPS	Free	TI 4100 NAVSTAR	1982	\$119,900	\$509,186
3 Digital voice recorder	Free	Sony PCM	1978	\$2,500	\$9,458
4 Digital watch	Free	Seiko 35SQ Astron	1969	\$1,250	\$8,402
5 10 MP camera	Free	Canon EOS-1Ds	2002	\$7,999	\$10,968
6 Scanner	Free	HP ScanJet Plus	1989	\$1,595	\$3,173
7 Video player	Free	Toshiba V-8000	1981	\$1,245	\$3,378
8 Video camera	Free	RCA CC010	1981	\$1,050	\$2,849
9 Music player	Free	Sony CDP-101 CD player	1982	\$900	\$2,301
10 Encyclopedia	Free	Compton's CD Encyclopedia	1989	\$750	\$1,492
11 Videogame console	Free	Atari 2600	1977	\$199	\$810
Total					\$1,191,056

* Year of launch

rk

Source: @rakwong

What is Different When Goods Become Digital?

- Physical goods: Atoms are costly to (re)produce
- Digital goods: Bits are cheap to reproduce
 - Free, perfect, instant
 - Zero transportation costs
- Implications
 - Costs shift to fixed rather than variable costs
 - Substantial economies of scale

What is Different When Goods Become Digital? Part 2

- Vast data collected by user interactions with our product
- How can we leverage this data to make better products?
- Data are great, but need principled analysis to understand and extract insights
 - Expertise is often the constraint, not data

What is Different When Goods Become Digital? Part 3

“If you are looking for a career where your services will be in high demand, you should find something where you provide a scarce, complementary service to something that is getting ubiquitous and cheap. So what’s getting ubiquitous and cheap? Data. And what is complementary to data? Analysis.”

- Hal Varian, Google Chief Economist

Question: What else is getting cheap? What is the scarce complement?

Digital Goods Are Easier to Modify / Reconfigure

Twenty Different Logos for Purdue



Question: How would you go about selecting the logo that sells best on a shirt in a physical store?

Question: How would you go about selecting the logo that sells best on a shirt in an online store?

Digital Goods Are Easy to Modify / Reconfigure

- It is often easy to randomize what version of a digital good a user interacts with
 - This is an A/B test
- We can then test empirically what version is best
- Harder to do in an offline environment
- In this class we will learn how to design and analyze A/B tests to improve our products

COURSE OVERVIEW

* Ulrich & Eppinger Chapter 3

High-Level Overview

- Digital products are fundamentally different from physical goods
 - But economic principles still apply
- There is no magic formula to digital product design
 - If I could teach you how to generate and execute on a billion dollar idea, I would be a billionaire myself (I am not)
- Focus instead on techniques to help you along in the process

What this Course is Not

- A product management course
 - A lot of the techniques we discuss are relevant to PMs
 - DevOps is a nice complement to this course
- A course on web design
 - Though we will learn how to evaluate our designs
- A course about designing physical goods using digital tools
 - This is not a course on CAD

In this Course We Will Learn:

- How to generate and assess new opportunities
- How to identify customer needs and measure what consumers value
- How to assess the competitive landscape and price our offering
- Launch and go-to market strategies
- How to assess the performance of our product
- How to continually modify and improve the product
- How to incorporate algorithms into digital products

Designing Algorithms

- Algorithms play an increasingly large role in digital products
 - Recommendation systems, personalization, automated decision making, human-AI collaboration, ...
- The last few lectures will be devoted to frameworks to design algorithms in digital products
- This will emphasize recent research on the topics as we actively try to understand such new tools

COURSE STRUCTURE

*** Ulrich & Eppinger Chapter 3**

Course Format

- Lectures: Tu/Th 4:30-5:45pm
- Participation is important
 - You all have unique experiences to share & contribute
- There are optional readings in the syllabus, class should be self-contained
 - Ulrich and Eppinger. Product Design and Development
 - Kohavi, Tang, and Xu. Trustworthy Online Controlled Experiments

Grading

- Participation (10%)
 - Come to class and contribute positively, hopefully everyone gets full credit here
 - Key is to not participate negatively

Grading, Part 2

- Individual assignments (40%)
 - Three individual assignments intended on reinforcing the concepts we learn in class. These will be empirical but can be done without any programming

Team Project (50%)

- Create a proposal for a digital product with a team of 4
- Apply the concepts we learned in the class to improve the proposal
- Example:
 - Apply survey techniques to learn what users value, set a price
 - Run an A/B test to refine a prototype
 - Do an analysis to refine the metrics you use to measure performance
 - ...

Team Project (50%), Continued

- Best projects come from things you care about
 - Business / community org / club you are involved in
 - Can also be hypothetical (i.e. a new product)
- Key Dates
 - 10/31: Team formed and product selected (email me your team)
 - 11/6: Midpoint check-ins
 - 12/9-12/11: Final project presentations + final report due

INTRODUCTIONS: ABOUT ME

Alex Moehring

Undergrad at UNC Chapel Hill

Industry Experience:
Wells Fargo Securities
Microsoft Research

PhD at MIT Sloan

Assistant Professor at Purdue Daniels



1. Ranking algorithms on social media
2. Human-AI collaboration
3. Social influence

All papers available at www.alexmoehring.com

Ranking Algorithms on Social Media

Personalization, Engagement, and Content Quality on Social Media: An Evaluation of Reddit's News Feed

Alex Moehring*

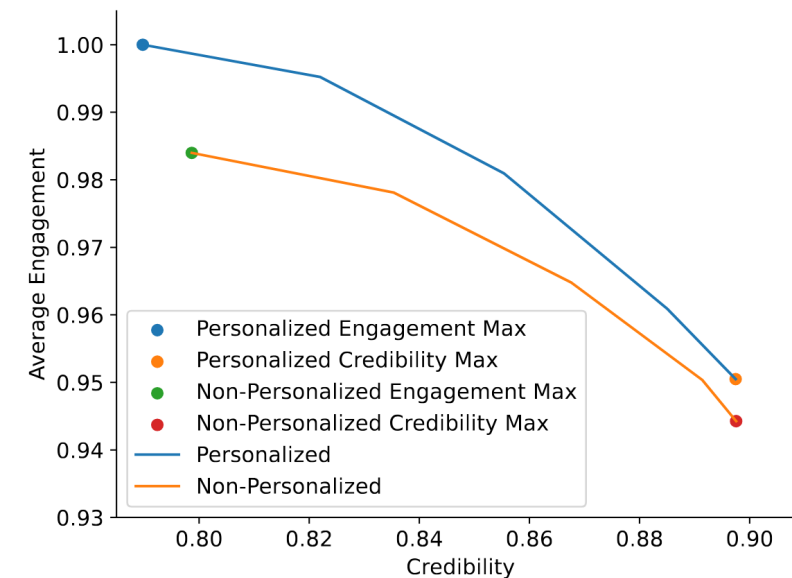
May 30, 2024

Abstract

Digital platforms increasingly curate their content through personalized algorithmic feeds. Platforms have an incentive to promote content that increases the predicted engagement of each user to lift advertising revenues. This paper studies how ranking content to maximize engagement affects the credibility of news content with which users engage. In addition, I evaluate how the ranking algorithm itself can be designed to promote engagement with high-credibility content. Using data from the Reddit politics community, I exploit a novel discontinuity in the ranking algorithm to identify the causal effect of a post's rank on the number of comments it receives. I use this discontinuity to identify a model of user comment decisions and estimate the credibility of news content that users engage with under a personalized engagement-maximizing algorithm. The personalized engagement-maximizing algorithm exacerbates differences in the credibility of news content with which users engage. I then evaluate a credibility-aware algorithm that explicitly promotes credible news publishers and find the platform can substantially increase the share of engagement with high-credibility publishers for a small reduction in total engagement. These findings suggest algorithmic interventions can be a useful tool for managers to balance engagement quantity and content quality.

$$d_{ijt} = 1[v_{ijt} = 1]1[u_{ijt} \geq u_{iot}]$$

Figure 7: Engagement-Quality Frontier



Human-AI Collaboration

Combining Human Expertise with Artificial Intelligence: Experimental Evidence from Radiology*

Nikhil Agarwal, Alex Moehring, Pranav Rajpurkar, Tobias Salz[†]

March 4, 2024

Abstract

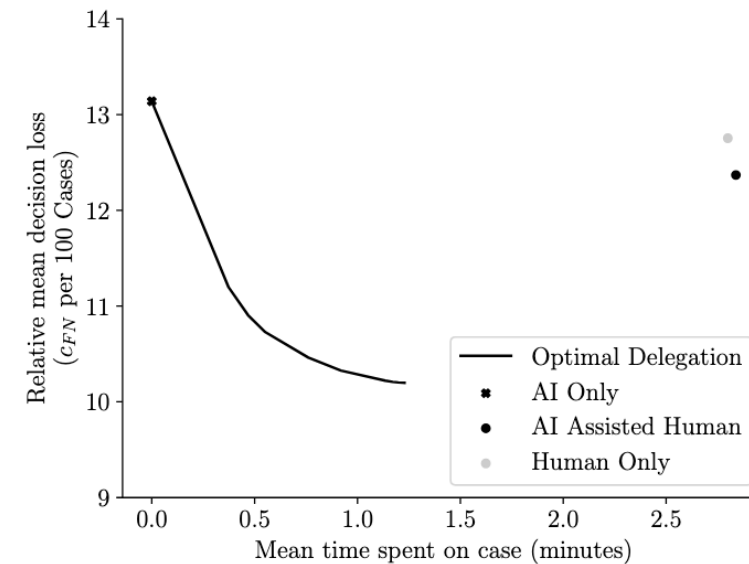
Full automation using Artificial Intelligence (AI) predictions may not be optimal if humans can access contextual information. We study human-AI collaboration using an information experiment with professional radiologists. Results show that providing (i) AI predictions does not always improve performance, whereas (ii) contextual information does. Radiologists do not realize the gains from AI assistance because of errors in belief updating – they underweight AI predictions and treat their own information and AI predictions as statistically independent. Unless these mistakes can be corrected, the optimal human-AI collaboration design delegates cases either to humans or to AI, but rarely to AI assisted humans.

JEL: C50, C90, D83, D47

Keywords: Artificial Intelligence, Human-AI Interaction, Belief Updating

$$u_h(a, \omega) = -1 \cdot \{a = 1, \omega = 0\} \cdot c_{FP,h} - 1 \cdot \{a = 0, \omega = 1\} \cdot c_{FN,h}.$$
$$a_h^*(s; p_h) = 1 \cdot \left\{ \frac{p_h(\omega = 1|s)}{p_h(\omega = 0|s)} > c_{rel,h} \equiv \frac{c_{FP,h}}{c_{FN,h}} \right\}.$$

Figure 10: Loss-time frontier



Additional Work

- Social influence online
 - How do social norms influence health behaviors
 - How do social image concerns impact online behaviors
- Ongoing work: Human-AI collaboration
 - How to design human-AI collaboration accounting for endogenous human responses

Office Hours + Contacting Me

- My office is Kran 540
- Office hours Tuesdays 6-7pm or by appointment
- Happy to discuss course content, advanced topics, applications, research, ...
- Email guidelines (moehring@purdue.edu)
 - MGMT 682: [[Informative summary]]
 - Follow-up if I don't respond within 1 business day

INTRODUCTIONS: ABOUT YOU

Introductions

- Name
- Program
- Why you took the course / what you hope to learn
- What industry you worked in previously or hope to work in after Purdue
- Project pitches: If you have a product you'd like to study in your project, make a pitch to classmates to join you!

QUESTIONS

Next Time

- Ideation & Opportunity Development